MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**
4000 PROCESS AREA: ADMIN BUILDING
DESIGN OF BUILDING STRUCTURE & FOUNDATION

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
Typ.	:	Typical
HCS	:	Hollow Core Slab
Fc'	:	Concrete Compressive Strength
Fy	:	Yield Strength of Reinforcing Steel
ϕ	:	Strength Reduction Factor
SBC	:	Saudi Building Code
SASO	:	Saudi Standards, Metrology and Quality Organization
ACI	:	American Concrete Institute
ASCE	:	American Society of Civil Engineers
PCI	:	Precast/Prestressed Concrete Institute

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1. INTENT

This Document is intended to produce the technical needs in Engineering including particular reference to design justification & Engineering validation works. The work described below is related to Design of Electrical building in Security Base Camp area.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The Scope of this document is limited to Complete Analysis & Structural Design Verification of Electrical Building only.

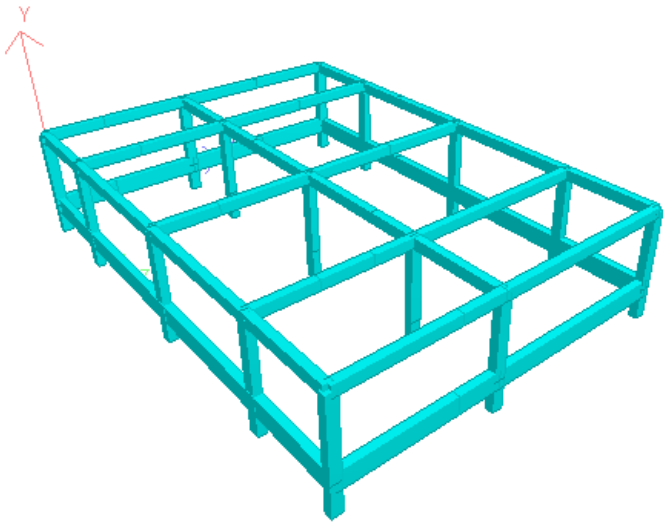
4. DESIGN METHODOLOGY & ASSUMPTIONS

This Building Is a Single Story Frame Structure with Foundation Being Cast in Site. The Building Shall Have Insulated Block Walls and Cast in Site Framing Beams to support the HCS (Hollow Core Slab)The Superstructure for this Building was modeled using STAAD software. The Analysis of Frames (Cast-In-Situ and Steel Columns & Beams) has been performed meeting design requirements & keeping in view the Architectural requirements. The loads were applied as per SBC/ASCE/ACI standards & analysis was done to generate the Column Reactions at the Base that are assumed to be fixed. The Foundation design has been done by importing Column Base Reactions into Foundation Design Software.

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STAAD 3D MODEL

4.1. Codes

[1] SBC-301-2007- Structural Loading & Forces.

[2] SBC-303-2007- Structural Soil & Foundations.

[3] SBC-304-2007- Structural Concrete Structures.

[4] SASO SSA 2 - Steel Bars for the Reinforcement of Concrete.

[5] SASO SSA 224 - Steel Fabric for Reinforcement of Concrete

[6] ASCE-7-2005 - Minimum Design Loads for Buildings and other Structures.

[7] ACI 318-08 - Building Code Requirements.

[8] PCI –MNL-120-04 - Design Handbook, 6th edition.

[9] SBC-306-2007- Structural – Steel Structures.

4.2. Geotechnical investigations

Foundations have been designed based on recommendations of the Soil Investigation Report, Doc # MD-216-3000-EG-CV-RPT-1008.

4.3. Material Properties

Compressive Strength of Concrete at 28th days

fc' = 35 MPa (5.00 Ksi) for beams, columns, suspended slabs.

fc' = 35 MPa (5.00 Ksi) for foundations, grade beams, columns.

Yield Strength of Steel

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fy = 414 MPa (60 Ksi) for all reinforcing steel.

5. DESIGN LOADS

5.1. Dead Load(DL)

5.1.1. Roof Dead Loads

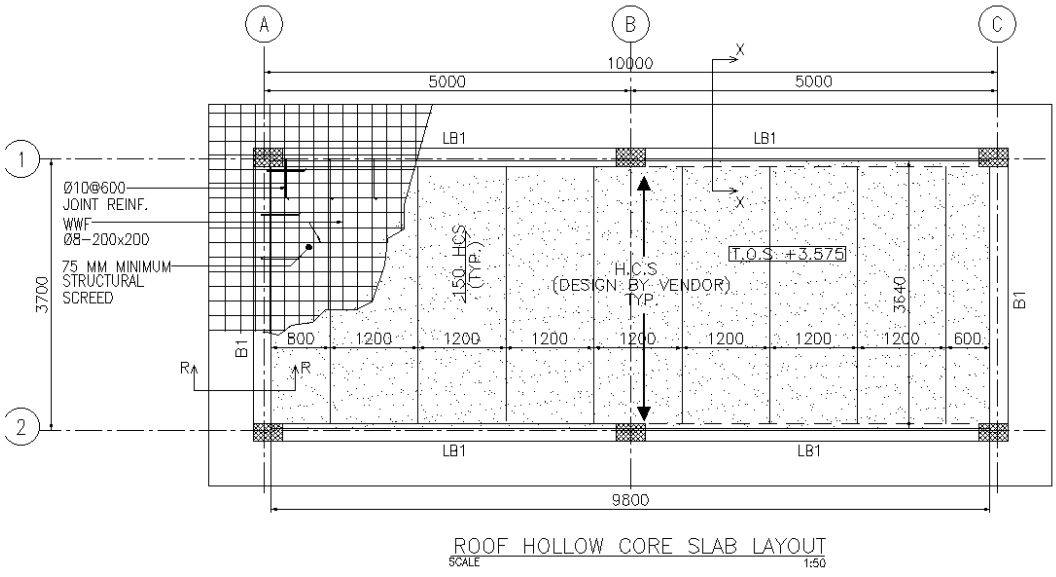
- Provision for Electrical Lighting = 0.05 kN/m²

Ceiling

Suspended steel channel system = 0.10 kN/m²

Acoustical Fiber Tile (16mm) = 0.05 kN/m²

Sub-total = **0.2 kN/m²**



ROOF HOLLOW CORE SLAB LAYOUT
SCALE 1:50

- Hollow Core Slab

(HCS-150)(To be confirmed by Vendor) b/w Grid Line's (1 & 4), (A & D) = **2.52 KN/m2**

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Prestressed Hollowcore Floor										
Overall Structural Depth (mm)	Self Wt- KN/M2	No. of Cores in Unit	Spans indicated below allow for Characteristic Service Load (Live Load) Plus Self Weight Plus 1.5Kn/m^2 for Finishes Characteristic Service Loads Kn/m^2							
			0.75	1.50	2.00	2.50	3.00	4.00	5.00	10.00
			Clear Span in metres							
100	2.38	0	5.39	5.19	5.08	4.91	4.72	4.40	4.14	3.30
150	2.52	11	7.40	7.40	7.40	7.40	7.28	6.80	6.40	5.12
150 (Dense	2.94	11	7.40	7.40	7.40	7.35	7.08	6.64	6.27	5.06
200	3.10	11	9.40	9.40	9.40	9.40	9.26	8.69	8.21	6.65
250	3.43	6	10.80	10.80	10.80	10.80	10.80	10.80	10.56	8.60
300	3.88	6	13.50	13.50	13.50	13.50	13.18	12.44	11.81	9.70
350	4.23	6	15.00	15.00	15.00	15.00	14.89	14.08	13.39	11.06
400	4.57	6	16.00	16.00	16.00	16.00	15.97	15.13	14.42	11.98
450	4.92	6	18.00	18.00	18.00	17.70	17.19	16.33	15.58	13.02
500	5.27	6	18.00	18.00	18.00	18.00	18.00	17.13	16.38	13.76

- Built-in Water Proofing System

100mm thk Polystyrene Foam

0.01 kN/m² per 25.4mm thickness x (100/25.4mm) = 0.039 kN/m²

Water proofing membrane

Bituminous, smooth surface = 0.07 kN/m²

90 mm thick average cement foam concrete screed Lightweight

Concrete 0.38 kN/m² per 25mm thickness x (90/25mm) = 1.368 kN/m²

75 mm Concrete screed above HCS = 21 x 0.075 =1.6 kN/m²

Sub-total = **3.1 kN/ m²**

Therefore

Total Dead Load =3.1+0.2+2.52= 5.82 **kN/m²**

TOTAL DL as UDL ON BEAMS

- UDL on beams due to Hollow Core Slab are calculated in table below.
- UDL on beams due to 325 mm high parapet = 0.325*24 = **7.8 kn/m**
- Moment on all the External Beams due to Cantilever Portion of slab and Screed = (0.15x24+ (0.05+0.2)/2x20) *0.6*0.6/2 + 0.325x24x0.6= **5.8 knm/m**

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Grid Mkd.	Span Left (m)	Span Right (m)	UDL (kN/m)
1	0	3.55	5.82 x (3.55+0)/2 = **10.4**
2	3.55	6.15	5.82 x (3.55+6.15)/2 = **28.3**
3	6.15	5.7	5.82 x (6.15+5.7)/2 = **34.5**
4	5.7	3.75	5.82 x (5.7+3.75)/2 = **27.5**
5	3.75	0	5.82 x (3.75+0)/2 = **11**

5.1.2. Ground Floor Walls

- 250 thk. Outer CMU (Partially Grouted Cell@800 c-c)

16 x 0.25 x 3.05m clear height = **12.2 kN/m**

5.1.3. Roof Live Loads (Lr)

Live loads are with reference to Clause-4.2.2, MD-216-1000-EG-ST-CRT-1001

Ordinary Flat (Inaccessible) = **0.5 kN/m²**

- UDL on external beams due to cantilever portion = 0.6 x 0.50 = **0.3 Kn/m**
- Moment on all the External Beams due to Live Load on Cantilever Portion of slab = 0.5*0.6*0.6/2 = **0.09 kN/m /m**
- UDL on grid beams due to roof live load are calculated in table below:

Grid Mkd.	Span Left (m)	Span Right (m)	UDL (kN/m)
1	0	3.55	0.5x (3.55+0)/2 = **0.9**
2	3.55	6.15	0.5 x (3.55+6.15)/2 = **2.5**
3	6.15	5.7	0.5 x (6.15+5.7)/2 = **3.0**
4	5.7	3.75	0.5 x (5.7+3.75)/2 = **2.4**
5	3.75	0	0.5 x (3.75+0)/2 = **1**

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5.2. Wind Load (WL)

Wind loads were automatically generated in STAAD as per ASCE-07. The parameters used for the wind load generation are as follows (Refer STRUCTURAL DESIGN CRITERIA NON-PROCESS BUILDINGS Doc No MD-216-0000-EG-ST-CRT-0001)

All structures and foundations (except for buried structures) designed and shall be constructed to resist the wind effects determined in accordance with Section-7.2, SBC-301-Analytical Procedure.

The Wind Loads generated in STAAD are in accordance with SBC-301-2007-Structural Loading & Forces.

Type of Roof	Flat
Length of building (X direction)	13.5 m
Width of building (Z direction)	19.5 m
Height of eaves	3.1 m
Mean height	3.1 m

General Wind Load parameters:

Basic wind speed:	V = 49 m/s (or) 177 kmph (Refer Clause-4.2.6 Doc-MD-216-1000-EG-ST-CRT-1001)
Risk category:	II (As per SBC-301, Table 1.6-1)
Velocity pressure coefficient:	K_d = **0.85** (As per SBC-301, Table 6.4-1)
Exposure category:	D (As per SBC-301, Section 6.4.2.3)
Enclosure classification:	Enclosed buildings
Gust effect factor:	G_f = **0.85** (As per SBC-301, Section-7.2.7.1)
Importance factor:	1 (As per SBC-301, Section-9.5)
Topography factor:	Kzt = 1.0 (As per SBC-301, Section 6.4.3.2

5.3. Seismic Load(E)

Earthquake loads were automatically generated in STAAD as per IBC 2003. The basic parameters affecting the generation of seismic loads is shown below:

Importance Factor, I = 1.0(As per SBC-301, Section-9.5)

Design Spectral Response Acceleration (Refer Clause-4.2.7b, MD-216-1000-EG-ST- CRT-1001)

Ss = 0.1g (Spectral Response Acceleration 0.2 second)

S1 = 0.02g (Spectral Response Acceleration 1 second)

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S_{D_1} = 2/3*Fv*S1 = 2/3 * 2.4 * 0.02 = 0.032 (As per SBC-301, Section-9.4.4)

S_{DS} = 2/3*Fa*Ss = 2/3 * 1.6 * 0.1 = 0.106 (As per SBC-301, Section-9.4.4)

Where S_{D_1} Damped Design Spectral Response Acceleration@1 Sec

S_{DS} Damped Design Spectral Response Acceleration@0.2 Sec

Site Class = D

Site Coefficient, Fa = 1.60 (As per SBC-301, Table 9.4.3a)

Site Coefficient, Fv = 2.40 (As per SBC-301, Table 9.4.3b)

Building period Coefficient, Ct=0.04 (As per SBC-301, Table 10.9.3.2)

For Ordinary reinforced concrete moment frame,

Response Modification Factor, Rx = 2.5 (As per SBC-301, Table 10.9.3.2)

Response Modification Factor, Rz = 2.5 (As per SBC-301, Table 10.2)

System Over Strength Factor, Omega = 3 (As per SBC-301, Table 10.2)

Deflection Amplification factor, Cd = 2.5 (As per SBC-301, Table 10.2)

5.4. Thermal/Temperature Loads(Self-restraining ,Thermal Expansion & Contraction Loads)

An ambient temperature ranges from a maximum of 50° C to a low of 0° C. In direct exposure to sunlight, the temperature shall be assumed to rise a minimum of 15° C above ambient. As a minimum, the temperature gradient for thermal loads shall be +25° C and -15° C with the most severe condition governing the design.

6. DESIGN COMBINATIONS

Service Loads (Allowable Stress Design) (SBC-301, Section 2.4.1)

D

D + T

D +Lr

D +0.75Lr

D + (W or 0.7E)

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D + 0.75W + 0.75Lr

0.6D + W

0.6D + 0.7E

Factored loads (Strength Design) (SBC-301, Section 2.3.2)

1.4D

1.4D+1.4T

1.2(D + T) + 0.5Lr

1.2D + 1.6Lr + 0.8W

1.2D + 1.6W + 0.5Lr

1.2D + 1.0E

0.9D + 1.6W

0.9D + 1.0E

Where:

D = Dead Load

Lr = Live Load

W = Wind Load

E = Seismic Load

T = Temperature Load

7. CRACKED SECTIONS

In lieu of a more accurate analysis of the effects of cracking of concrete, provisions of SBC-304-Section 10.11 was used.

For Beams: I = 0.35Ig

For Columns: I = 0.70Ig

For Walls:

Un-cracked I = 0.70Ig

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Cracked I = 0.35Ig

For Slabs I = 0.25Ig

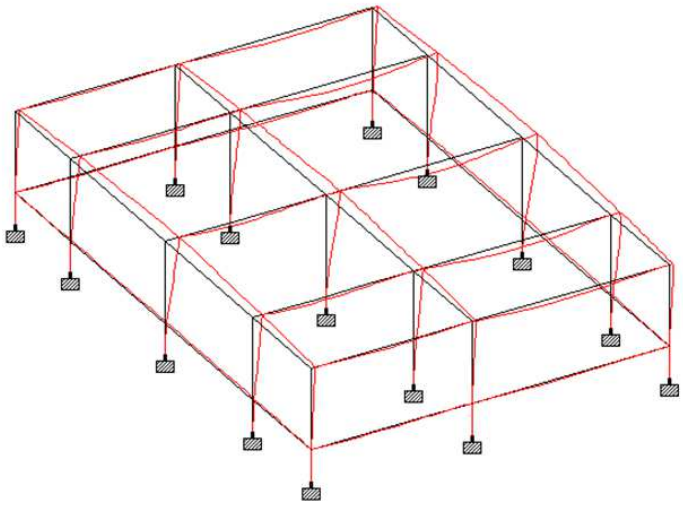
8. DEFLECTIONS

8.1. Maximum Allowable Deflection

L/240: for member load with live load plus dead load

L/600: for members supporting masonry under total loads

Where, L = span of member between support



All / Summary

	Node	L/C	Horizontal X m	Vertical Y m	Horizontal Z m	Resultant m
Max X	54	118 0.6D+0.7	**0.011**	-0.003	0.000	0.011
Min X	61	108 D+0.7EZ	**-0.001**	-0.002	0.003	0.004
Max Y	9	101 DL+T	0.000	**0.000**	0.000	0.000
Min Y	60	102 DL+LR	-0.001	**-0.014**	0.000	0.014
Max Z	60	108 D+0.7EZ	-0.001	-0.013	**0.009**	0.016
Min Z	52	118 0.6D+0.7	0.003	-0.002	**-0.000**	0.004
Max rX	57	108 D+0.7EZ	-0.000	-0.001	0.007	0.007
Min rX	58	102 DL+LR	-0.001	-0.007	0.000	0.007
Max rY	24	118 0.6D+0.7	0.008	-0.000	-0.000	0.008
Min rY	55	108 D+0.7EZ	-0.000	-0.003	0.006	0.007
Max rZ	33	102 DL+LR	-0.001	-0.000	0.000	0.001
Min rZ	32	109 D+0.7EX	0.003	-0.000	-0.000	0.003
Max Rs	60	108 D+0.7EZ	-0.001	-0.013	0.009	**0.016**

Maximum attained Drift as per Analysis from above figure's = **14 mm**

Allowable Drift as per Table-10.12, SBC-301-2007 is H/100 = 3000/100 = **30mm**

Hence the Structure is safe considering drift check…

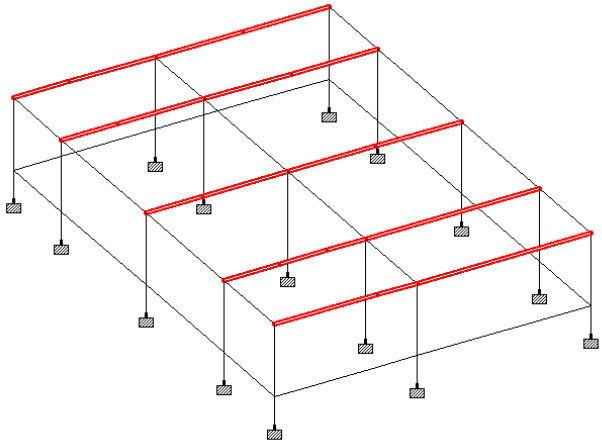
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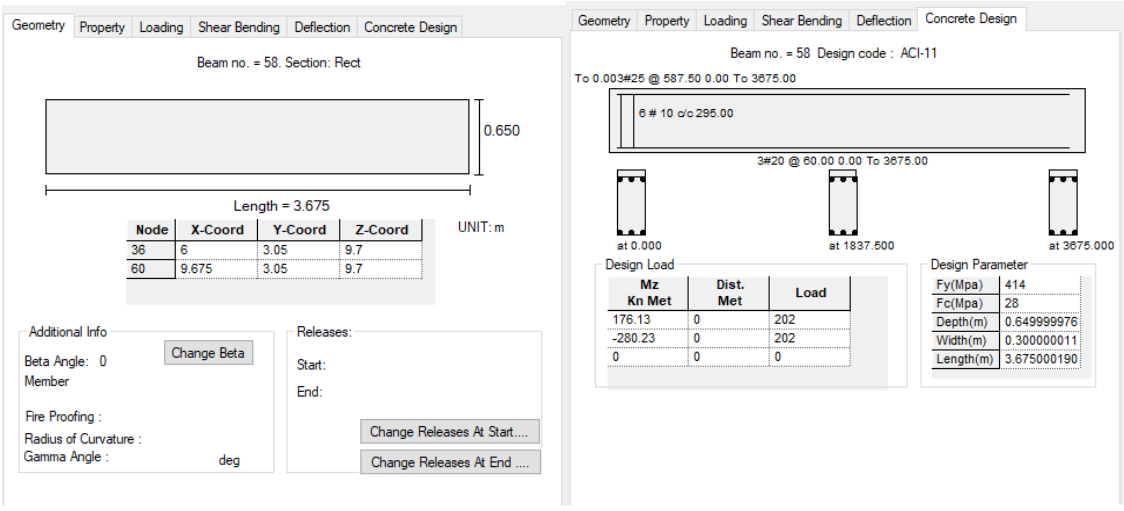
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9. CONCRETE ROOF BEAM DESIGN

9.1. Design of Ledger Beam(LB1)





Design of Ledger Beam (LB1)

Maximum Moment required from Analysis = 280.5 **Knm (Hogging)**

Therefore Ast.req = $$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

Where b=250mm, Fy=414 Mpa, d=650-40-10-25/2=587.5 mm, Fc'=35 Mpa & ϕ =0.9

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Ast.req = **1375 mm²**

Ast.min=
(As per clause-10.5-ACI-318)

$$A_{s,\,min} = \frac{0.25\sqrt{f'_c}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **564mm²**

Hence Provide 3-25 dia-Full Straight Bars Top & Bottom

Ast.pro= **1470mm² >>>>>>(Ast.min, Ast.req)**

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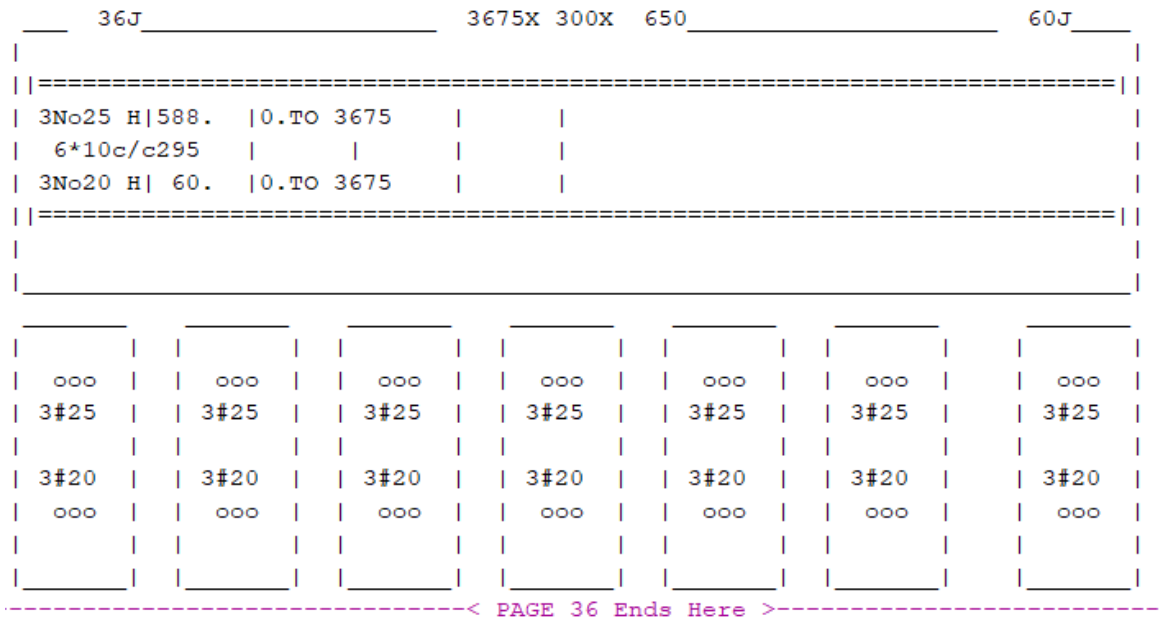
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9.1.1. Staad Output for LB1

```
            B E A M   N O.    58 D E S I G N   R E S U L T S - SHEAR


   AT START SUPPORT - Vu=  194.33 KNS  Vc=  158.82 KNS  Vs=  100.29 KNS
   Tu=    0.14 KN-MET  Tc=    6.8 KN-MET  Ts=    0.0 KN-MET  LOAD   202
   NO STIRRUPS ARE REQUIRED FOR TORSION.
   REINFORCEMENT IS REQUIRED FOR SHEAR.
   PROVIDE 10 MM 2-LEGGED STIRRUPS AT 295. MM  C/C FOR 1248. MM

   AT END   SUPPORT - Vu=   38.53 KNS  Vc=  150.92 KNS  Vs=    0.00 KNS
   Tu=    0.07 KN-MET  Tc=    6.7 KN-MET  Ts=    0.0 KN-MET  LOAD     3
                    STIRRUPS ARE NOT REQUIRED.
```



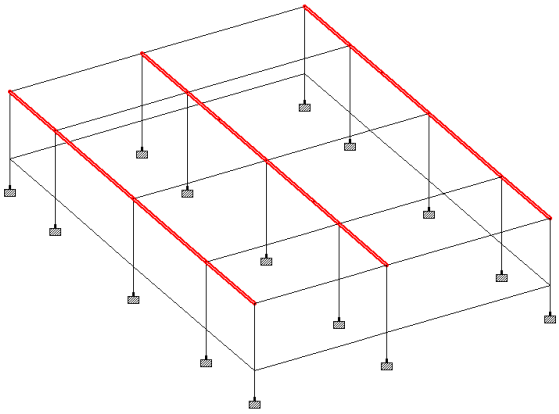

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   STAAD SPACE                                          -- PAGE NO.    37
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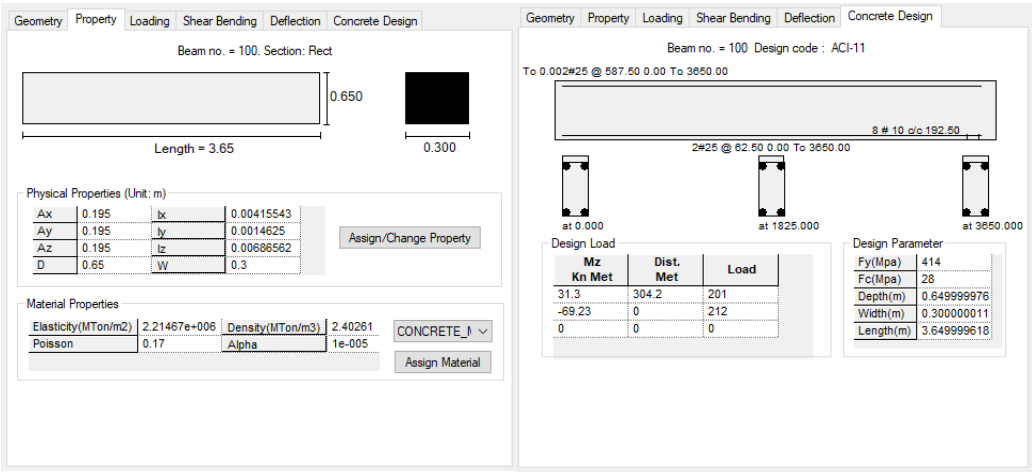

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9.2. Design of Beam(B1)





Hence Provide 2-25 dia-Full Straight Bars Top & Bottom

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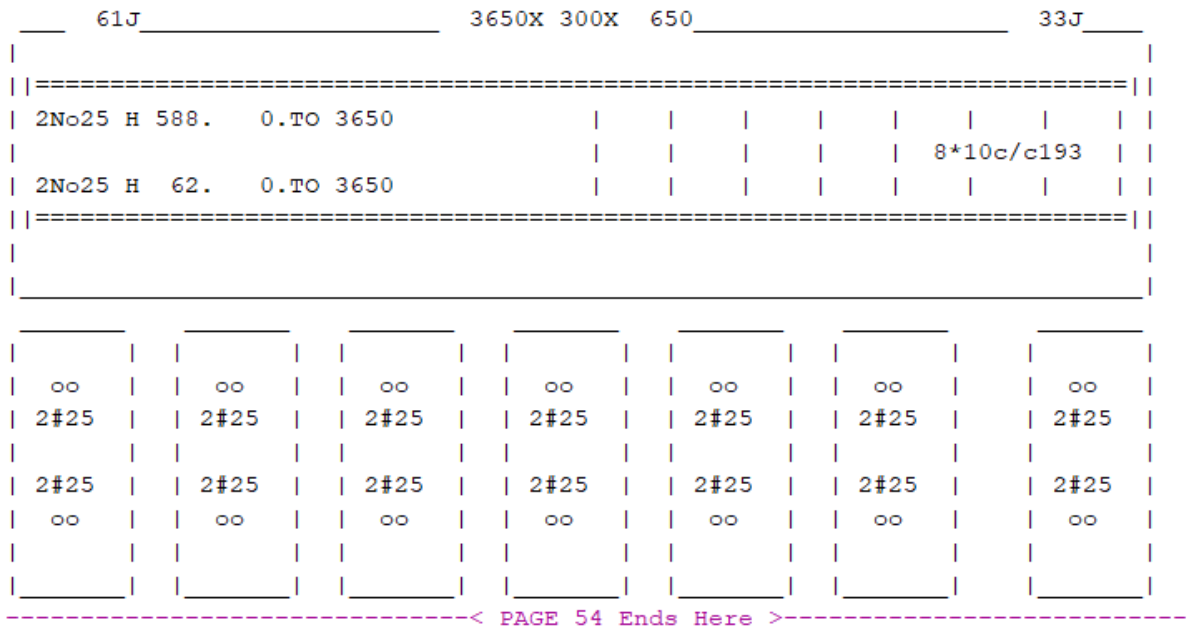
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9.2.1. Staad Output for B1

```
            B E A M  N O.   100 D E S I G N  R E S U L T S - SHEAR


   AT START SUPPORT - Vu=     1.77 KNS   Vc=  147.30 KNS   Vs=     0.00 KNS
   Tu=     1.54 KN-MET   Tc=     6.7 KN-MET   Ts=     0.0 KN-MET   LOAD      3
                     STIRRUPS ARE NOT REQUIRED.
   AT END   SUPPORT - Vu=    44.34 KNS   Vc=  152.33 KNS   Vs=     0.00 KNS
   Tu=    22.47 KN-MET   Tc=     7.0 KN-MET   Ts=    30.0 KN-MET   LOAD    202
   STIRRUPS ARE REQUIRED FOR TORSION.
   REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
   PROVIDE 10 MM 2-LEGGED STIRRUPS AT 193. MM  C/C FOR 1235. MM
   ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =    5.57 SQ.CM.
```




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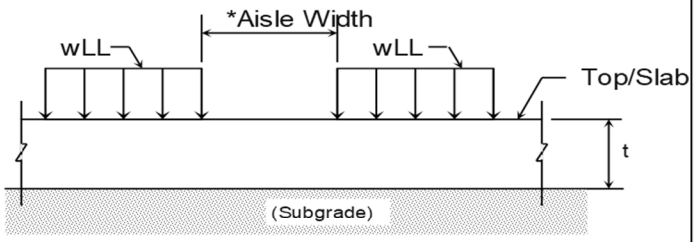
10. CONCRETE GRADE SLAB DESIGN

CONCRETE SLAB ON GRADE ANALYSIS

For Slab Subjected to Stationary Uniformly Distributed Live Loads

Input Data:

Slab Thickness, t =	3.937	in.	100.000	mm		
Concrete Strength, f 'c =	5000	psi	35.000	kpa		
Subgrade Modulus, k =	39.7866	pci	10800	Kn/m3	(AS PER SECTION-7.7-SOIL REPORT)	
Factor of Safety, FS =	1.500		1.500			
Uniform Live Load, wLL =	104.45	psf	5.00	kpa		



Concrete Slab on Grade with Uniform Loads

Results:

Design Parameters:

Modulus of Rupture, MR =	636.40	psi	4387.79	kpa	MR = 9*SQRT(f 'c)
Allow. Bending Stress, Fb =	424.26	psi	2925.19	kpa	Fb = MR/FS
Modulus of Elasticity, Ec =	4030509	psi	27789329	kpa	Ec = 57000*SQRT(f 'c)
Poisson's Ratio, μ =	0.15		0.15		μ = 0.15 (assumed for concrete)
Radius of Stiffness, Lr =	26.94	in.	684.37	mm	Lr = (Ec*t^3/(12*(1-μ^2)*k))^0.25
Critical Aisle Width, Wcr =	4.96	ft.	1511.77	mm	Wcr = (2.209*Lr)/12

Stationary Uniformly Distributed Live Loads:

wLL(allow) =	682.05	psf	wLL(allow) = 257.876*Fb*SQRT(k*t/Ec)
	32.65	kpa	**wLL(allow) >= wLL, O.K.**

Provide Minimum Reinforcement of 0.14%@both faces = 0.14%(h) = 140 mm2/M

Table 24.4.3.2—Minimum ratios of deformed shrinkage and temperature reinforcement area to gross concrete area

Reinforcement type	f_y, psi	Minimum reinforcement ratio	
Deformed bars	< 60,000	0.0020	
Deformed bars or welded wire reinforcement	≥ 60,000	Greater of:	$\frac{0.0018 \times 60,000}{f_y}$
			0.0014

Provide WWF-8mm Dia@200 c-c Both Ways-TOP LAYER

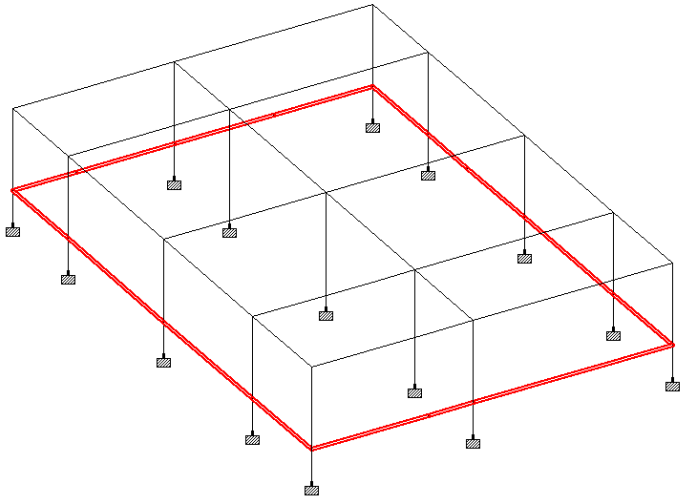
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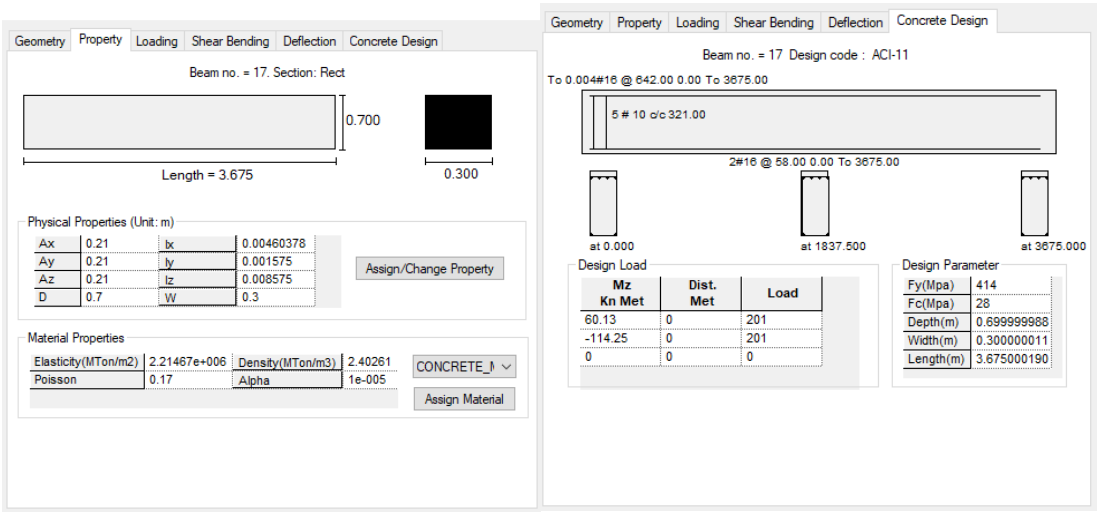
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11. DESIGN OF GRADE BEAM

11.1. Grade Beam(GB1)





Hence Provide 4-16 dia-Full Straight Bars Top & 3-16 dia-Full Straight Bars Bottom

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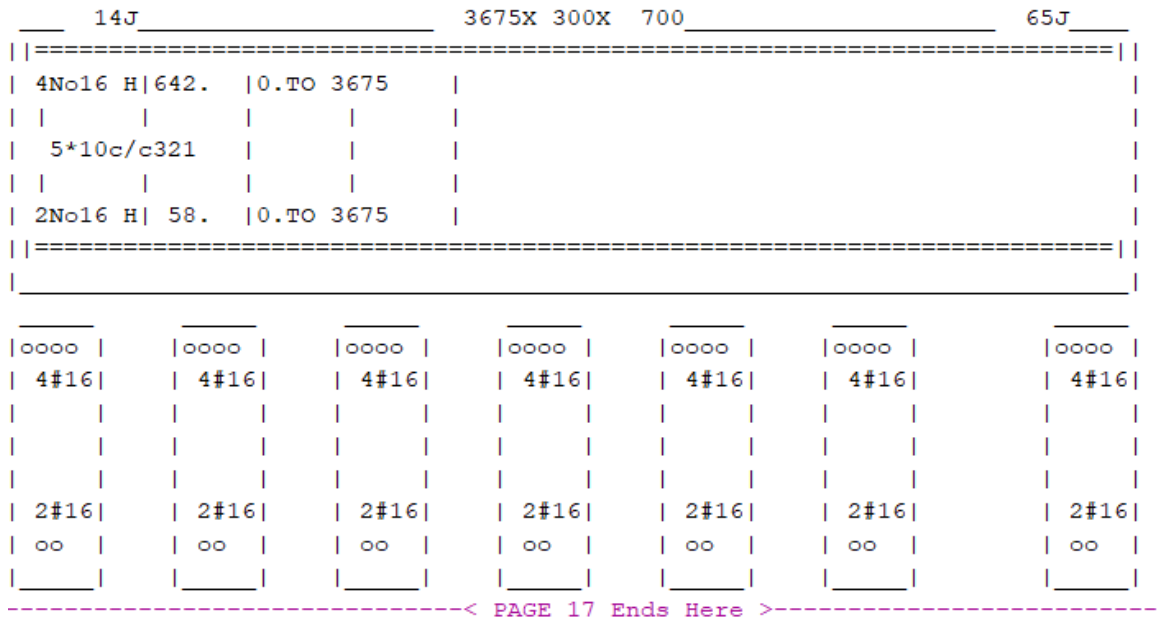
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11.1.1. Staad Output for GB1

```
            B E A M  N O.    17 D E S I G N  R E S U L T S - SHEAR


   AT START SUPPORT - Vu=   76.20 KNS  Vc=  165.51 KNS  Vs=    0.00 KNS
   Tu=    0.11 KN-MET  Tc=    7.3 KN-MET  Ts=    0.0 KN-MET  LOAD   201
   NO STIRRUPS ARE REQUIRED FOR TORSION.
   REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
   PROVIDE 10 MM 2-LEGGED STIRRUPS AT 321. MM  C/C FOR 1198. MM

   AT END   SUPPORT - Vu=   13.36 KNS  Vc=  162.11 KNS  Vs=    0.00 KNS
   Tu=    0.08 KN-MET  Tc=    7.3 KN-MET  Ts=    0.0 KN-MET  LOAD     3
                    STIRRUPS ARE NOT REQUIRED.
```




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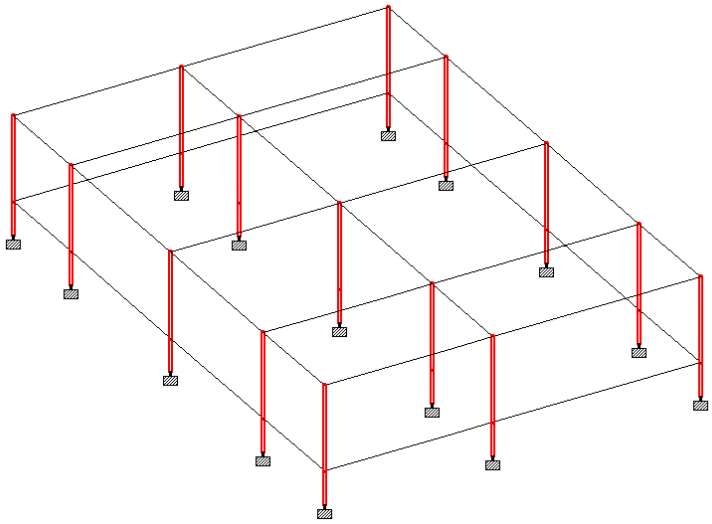

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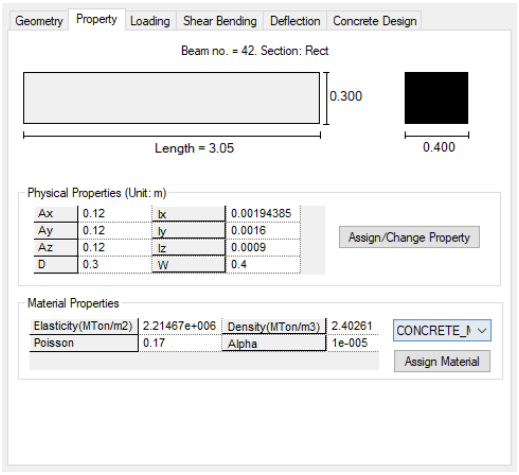
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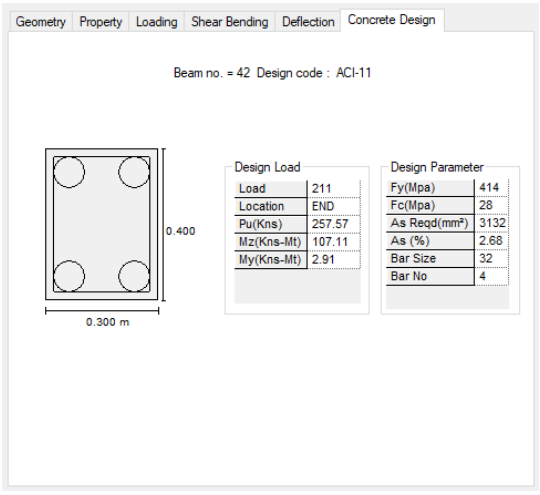
12. CONCRETE COLUMN DESIGN

12.1. Design of Column-(C1)



Column C1





Hence Provide 8-25 dia-bar, Ast,provided = 3927 mm2.

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12.1.1. Staad Output for Column C1

```
ACI 318-11   COLUMN  NO.    42   DESIGN RESULTS
==================================================

FY - 414.0  FC -  28.0 MPA,  RECT SIZE - 400.0 X 300.0 MMS, TIED
        AREA OF STEEL REQUIRED =  3132.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
------------------------------------------------------------

  4 - 32 MM               2.681      211      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 300.00 MM
```


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13. CANTILEVER SLAB DESIGN

DESIGN OF CANTILEVER SLAB

Geometrical Properties			Material Properties		
Span to center of support "sp_1"	0.8	met	Weight of reinforced concrete "γ_r	24	kN/m^3
Clear Span	0.6	met	Strength of concrete "f'_c"	35	N/mm^2
Clear span + depth of slab "sp_2"	0.75	met	Strength of steel "f_y"	420	N/mm^2
Max. eff span le = min (sp_1;sp_2)	0.8	met	Clear cover to concrete("c")	20	mm
Width of Slab "b"	1000	mm	Flexure strength reduction factor	0.9	
Thickness of slab "t"	150	mm	Shear strength reduction factor	0.75	
Eff. Depth "d"	125		Modulus of Elasticity of steel "Es'	2E+05	N/mm^2
Loading			Modulus of Elasticity of concrete	27806	N/mm^2
Self Weight of slab	3.6	kN/m^2	Modular Ratio "n = Es/Ec"	7.19	
Floor Finish	3.5	kN/m^2	**Provided Reinforcement**		
Wall load	0	kN/m^2	Top layer Nos.	6 Φ 10	mm
Total dead Load "U_{dl}"	7.1	kN/m^2	Bottom layer Nos.	0 Φ 12	mm
Live load	1	kN/m^2	Area of Top steel "$A_{s\ t}$"	468	mm^2
Total Ultimate Load "$Uu_{,tot}$"$_{(1.2D+1.6L)}$	10.12	kN/m^2	Area of Bottom steel "$A_{s\ b}$"	0	mm^2
Moment due to Overhang Point Load(0.125*0.15*24*0.6)	0.27	kNm	"d_b' = 10 mm		

Calculations

Shear force Vu = U*le	**8.096**	**kN**
Neg. Moment Mu = U*le^2/2	**3.51**	**kNm**

Check for Section capacity

Depth of comp. block "a" = $A_{s\ t}$*f_y/(0.85*f'_c*b)	6.61	mm						
Section capacity ΦMn = Φ*$A_{s\ t}$*f_y*(d-a/2)	**21.94**	**kNm**	**O.K**	>	**3.51**	**kNm**	(Mu)	

Shear requirements

Shear capacity ΦVc = Φ*0.17*√f'c*b*d	**94.29**	**kN**	**O.K**	>	**8.096**	**kN**	(Vu)	

Serviceability requirements

d' = c+d_b/2	26	mm
B = b/(n*$A_{s\ t}$)	0.297	
r = (n-1)*$A_{s\ b}$/(n*$A_{s\ t}$)	0	
k_d = (√(2*d*B(1+r*d'/d)+(1+r)^2)-(1+r))/B	29.01	mm
I_{cr} = bkd^3/3+n*$A_{s\ t}$*(d-k_d)^2+(n-1)*$A_{s\ b}$*(k_d-d'	3.915E+07	mm^4
Calculate Gross moment of inertia		
Area of slab A = t*b	2E+05	mm^2
Depth of neutral axis "y"	75	mm
Distance of top fiber from N.A "y_t"	75	mm
Gross moment of inertia "I_g"	2.813E+08	mm^4
Allowable stress fr = 0.62√f'c	3.67	mm^2
Cracking moment M_{cr} = fr*I_g/y_t	13.75	kNm

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Service moments

Moment due to dead load "M_{dl}"	2.54	kNm	
Moment due to live load "Mll"	0.32	kNm	
Total Moment Ms = $M_{dl}+M_{ll}$	2.86	kNm	
Deflection due to dead + live loads			
R(dl+ll) = M_{cr}/Ms	1.000		
$I_e(dl+ll) = R^3*I_g+(1-R^3)*I_{cr}$	2.813E+08	mm^4	
$\delta = K*5/48*Ms*le^2/(Ec*I_e)$	0.08	mm	**K = 3.2 (conservative)**

Deflection due to dead load

R(dl) = M_{cr}/Md	5.411		
$I_e(dl) = R^3*I_g+(1-R^3)*I_{cr}$	3.839E+10	mm^4	
$\delta = K*5/48*Md*le^2/(Ec*I_e)$	0.00	mm	

Deflection due to live load

Live load deflection = dl+ll deflection - dl deflection 0.08 mm

Allowable immediate deflection due to Live load = per ACI 318 Tab 9.5b **$I_e/360$ 2.22 mm > 0.08 mm O.K**

Long term deflection

% of liveload sustained "SL"	50	%	
$M_{sus} = M_{dl}+SL*M_{ll}$	2.70	kNm	
R(sus) = $M_{cr}/Msus$	5.091		
$I_e(sus) = R^3*I_g+(1-R^3)*I_{cr}$	3.198E+10		
$\delta = K*5/48*Msus*le^2/(Ec*I_e)$	0.00	mm	

Magnification factor for creep and shrinkage

$\lambda_\Delta = \xi / (1+50*\rho')$ 2.000

Where:-

ξ = 2 for 5 years or more cl.9.5.2.5 ACI 318

$\rho' = A_{s\ b}/(b*d)$ 0

Total deflection = λ_Δ*sus deflection + ll deflection 0.079 mm

Allowable deflection = $I_e/240$ 3.333 mm > 0.079 mm O.K

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14. FOUNDATION DESIGN

Foundations have been designed based on the Soil Report Recommendations. The minimum Embedment Depth of Foundations has been considered as 1.2m & applicable SBC of 350 Kpa. (Refer Doc-MD-216-3000-EG-CV-RPT-1008 for Details)

TABLE 4: RECOMMENDED NET ALLOWABLE BEARING CAPACITIES

PRPJECT: MANSOURAH - MANSSARAH GOLD PROJECT
CLIENT: L & T CONSTRUCTION

Profile / Borehole	Foundation Type	Depth of Foundation below FGL	Footing Size (B x L)	Net Allowable Bearing Capacity Based on		Recommended Net Allowable Bearing Capacity	Estimated Total Settlement
				Settlement	Shear Failure		
-	-	(m)	(m)	(kN/m^2)	(kN/m^2)	(kN/m^2)	(mm)
BH-01	Square Isolated	1.20	1.0 x 1.0	1500	570	570	25
		1.50		2000	850	850	25
		1.20	2.0 x 2.0	1000	750	750	25
		1.50		1200	980	980	25
		1.20	3.0 x 3.0	775	900	775	25
		1.50		950	1120	950	25
		1.20	4.0 x 4.0	625	1040	625	25
		1.50		750	1250	750	25
		1.20	5.0 x 5.0	525	1172	525	25
		1.50		600	1342	600	25

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DESIGN CALCULATION OF FOUNDATION FOR ADMIN BUILDING

	Along-X	Along-Z
Size of column :	0.30m	0.30m
Size of fdn. raft :	1.50m	1.50m
Thickness of raft :	0.40	m
Original ground level:	0.00	m
Finished floor level :	0.50	m
U/S level of foundation :	-1.20	m
U/S level Base Plate :	0.15	m
Unit weight of conc :	24	kN/m³
Unit weight of soil :	18	kN/m³

SYMBOLS AND NOTATIONS

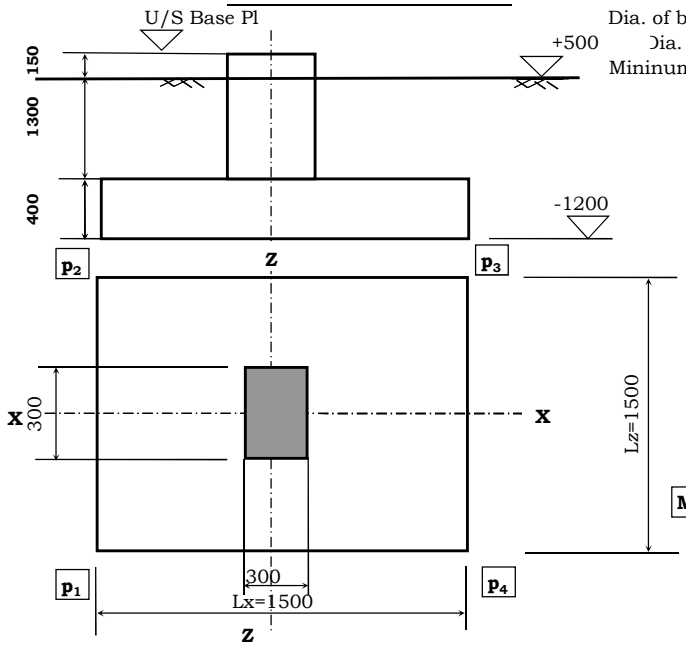
Dead Load	DL
Normal Operational Fluid Load	FL1
Max. Fluid Load	FL2
Wind Load	WL
Seismic Load	SL

Calculation of Weight of Foundation :

ght of raft & Col.=	(1.5*1.5*0.4*24)+(0.3*0.3*0.95*24) =	23.65KN
Weight of soil =	((1.5*1.5)-(0.3*0.3))*1.3*18 =	50.54KN
	Total =	74.20KN

Design Input:

Characteristic strength of concrete, f'_c=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	16mm
Dia. of bars provd. along X direction at top =	16mm
Dia. of bars provided along Z direction at bottom=	16mm
Dia. of bars provided along Z direction at top=	16mm
Mininum percentage of reinforcement at bottom=	0.18%

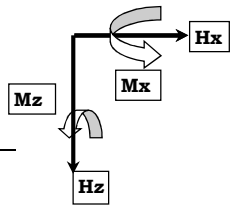


Effective depth of slab along X direction = 342mm = 0.342m

Effective depth of slab along Z direction = 326mm = 0.326m

Clear projection along Z direction = 0.6m

Clear projection along X direction = 0.6m



H_X=Force along X-X direction

Hz=Force along Z-Z direction

M_X=Moment about X-X direction

Mz=Moment about Z-Z direction

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Sensitivity: LNT Construction Internal Use

DESIGN CALCULATION OF FOUNDATION FOR ADMIN BUILDING

Combined Load Cases:	**Load at bottom of foundation**					**Load at bottom of foundation + self weight**				
Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design										
10 DL	340.8	6.3	-10.9	-0.3	-0.6	415.02	6.35	-10.89	-0.33	-0.64
101 DL+T	340.8	6.3	-10.9	-0.3	-0.6	415.02	6.35	-10.89	-0.33	-0.64
102 DL+LR	364.3	6.8	-11.7	-0.3	-0.6	438.53	6.84	-11.73	-0.34	-0.65
103 D+0.75LR	358.5	6.7	-11.5	-0.3	-0.6	432.65	6.71	-11.52	-0.34	-0.65
104 D+WX1	340.8	6.3	-10.9	-0.3	-0.6	415.02	6.35	-10.89	-0.33	-0.64
105 D+WX2	340.8	6.3	-10.9	-0.3	-0.6	415.02	6.35	-10.89	-0.33	-0.64
106 D+WZ1	340.8	6.3	-10.9	-0.3	-0.6	415.02	6.35	-10.89	-0.33	-0.64
107 D+WZ2	340.8	6.3	-10.9	-0.3	-0.6	415.02	6.35	-10.89	-0.33	-0.64
108 D+0.7EZ	341.8	6.3	-10.9	-21.4	-51.3	416.05	6.34	-10.87	-21.40	-51.26
109 D+0.7EX	343.3	-14.5	37.2	-0.3	-0.6	417.47	-14.51	37.18	-0.31	-0.59
110 D+0.75WX1+0.75LR	358.5	6.7	-11.5	-0.3	-0.6	432.65	6.71	-11.52	-0.34	-0.65
111 D+0.75WX2+0.75LR	358.5	6.7	-11.5	-0.3	-0.6	432.65	6.71	-11.52	-0.34	-0.65
112 D+0.75WZ1+0.75LR	358.5	6.7	-11.5	-0.3	-0.6	432.65	6.71	-11.52	-0.34	-0.65
113 D+0.75WZ2+0.75LR	358.5	6.7	-11.5	-0.3	-0.6	432.65	6.71	-11.52	-0.34	-0.65
114 0.6D+WX1	204.5	3.8	-6.5	-0.2	-0.4	278.69	3.81	-6.53	-0.20	-0.38
115 0.6D+WX2	204.5	3.8	-6.5	-0.2	-0.4	278.69	3.81	-6.53	-0.20	-0.38
116 0.6D+WZ1	204.5	3.8	-6.5	-0.2	-0.4	278.69	3.81	-6.53	-0.20	-0.38
117 0.6D+WZ2	204.5	3.8	-6.5	-0.2	-0.4	278.69	3.81	-6.53	-0.20	-0.38
118 0.6D+0.7EX	206.9	-17.0	41.5	-0.2	-0.3	281.14	-17.04	41.54	-0.18	-0.33
119 0.6D+0.7EZ	205.5	3.8	-6.5	-21.3	-51.0	279.72	3.80	-6.51	-21.27	-51.00

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Sensitivity: LNT Construction Internal Use

Combined Load Cases:	**Load at bottom of foundation**					**Load at bottom of foundation + self weight**				
Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Strength Design										
201 1.4DL	477.1	8.9	-15.2	-0.5	-0.9	551.35	8.89	-15.25	-0.46	-0.89
202 1.4D+1.4T+0.5LR	484.8	9.3	-16.0	-1.0	-2.1	559.01	9.28	-16.03	-0.95	-2.15
203 1.2D+0.8WX1+1.6LR	446.6	8.4	-14.4	-0.4	-0.8	520.80	8.40	-14.41	-0.41	-0.78
204 1.2D+0.8WX2+1.6LR	446.6	8.4	-14.4	-0.4	-0.8	520.80	8.40	-14.41	-0.41	-0.78
205 1.2D+0.8WZ1+1.6LR	446.6	8.4	-14.4	-0.4	-0.8	520.80	8.40	-14.41	-0.41	-0.78
206 1.2D+0.8WZ2+1.6LR	446.6	8.4	-14.4	-0.4	-0.8	520.80	8.40	-14.41	-0.41	-0.78
207 1.2D+1.6WX1+0.5LR	420.7	7.9	-13.5	-0.4	-0.8	494.94	7.86	-13.49	-0.40	-0.77
208 1.2D+1.6WZ1+0.5LR	420.7	7.9	-13.5	-0.4	-0.8	494.94	7.86	-13.49	-0.40	-0.77
209 1.2D+1.6WX2+0.5LR	420.7	7.9	-13.5	-0.4	-0.8	494.94	7.86	-13.49	-0.40	-0.77
210 1.2D+1.6WZ2+0.5LR	420.7	7.9	-13.5	-0.4	-0.8	494.94	7.86	-13.49	-0.40	-0.77
211 1.2D+EX	412.5	-22.2	55.6	-0.4	-0.7	486.68	-22.17	55.61	-0.37	-0.69
212 1.2D+EZ	410.5	7.6	-13.0	-30.5	-73.1	484.65	7.61	-13.04	-30.50	-73.08
213 0.9D+1.6WX1	306.7	5.7	-9.8	-0.3	-0.6	380.94	5.71	-9.80	-0.30	-0.57
214 0.9D+1.6WZ1	306.7	5.7	-9.8	-0.3	-0.6	380.94	5.71	-9.80	-0.30	-0.57
215 0.9D+1.6WX2	306.7	5.7	-9.8	-0.3	-0.6	380.94	5.71	-9.80	-0.30	-0.57
216 0.9D+1.6WZ2	306.7	5.7	-9.8	-0.3	-0.6	380.94	5.71	-9.80	-0.30	-0.57
217 0.9D+EZ	308.2	5.7	-9.8	-30.4	-72.9	382.41	5.70	-9.77	-30.40	-72.89
218 0.9D+EX	310.2	-24.1	58.9	-0.3	-0.5	384.43	-24.08	58.87	-0.27	-0.50

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Sensitivity: LNT Construction Internal Use

Check for overturning and sliding:

						Overturning		Sliding	
Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
10 DL	415.0	6.3	-10.9	-0.3	-0.6	28.6	489.4	26.2	503.1
101 DL+T	415.0	6.3	-10.9	-0.3	-0.6	28.6	489.4	26.2	503.1
102 DL+LR	438.5	6.8	-11.7	-0.3	-0.6	28.0	506.8	25.7	522.1
103 D+0.75LR	432.7	6.7	-11.5	-0.3	-0.6	28.2	502.3	25.8	516.6
104 D+WX1	415.0	6.3	-10.9	-0.3	-0.6	28.6	489.4	26.2	503.1
105 D+WX2	415.0	6.3	-10.9	-0.3	-0.6	28.6	489.4	26.2	503.1
106 D+WZ1	415.0	6.3	-10.9	-0.3	-0.6	28.6	489.4	26.2	503.1
107 D+WZ2	415.0	6.3	-10.9	-0.3	-0.6	28.6	489.4	26.2	503.1
108 D+0.7EZ	416.0	6.3	-10.9	-21.4	-51.3	28.7	6.1	26.3	7.8
109 D+0.7EX	417.5	-14.5	37.2	-0.3	-0.6	8.4	534.3	11.5	540.4
110 D+0.75WX1+0.75LR	432.7	6.7	-11.5	-0.3	-0.6	28.2	502.3	25.8	516.6
111 D+0.75WX2+0.75LR	432.7	6.7	-11.5	-0.3	-0.6	28.2	502.3	25.8	516.6
112 D+0.75WZ1+0.75LR	432.7	6.7	-11.5	-0.3	-0.6	28.2	502.3	25.8	516.6
113 D+0.75WZ2+0.75LR	432.7	6.7	-11.5	-0.3	-0.6	28.2	502.3	25.8	516.6
114 0.6D+WX1	278.7	3.8	-6.5	-0.2	-0.4	32.0	548.6	29.3	563.0
115 0.6D+WX2	278.7	3.8	-6.5	-0.2	-0.4	32.0	548.6	29.3	563.0
116 0.6D+WZ1	278.7	3.8	-6.5	-0.2	-0.4	32.0	548.6	29.3	563.0
117 0.6D+WZ2	278.7	3.8	-6.5	-0.2	-0.4	32.0	548.6	29.3	563.0
118 0.6D+0.7EX	281.1	-17.0	41.5	-0.2	-0.3	5.1	635.1	6.6	635.3
119 0.6D+0.7EZ	279.7	3.8	-6.5	-21.3	-51.0	32.2	4.1	29.4	5.3
					Min.=	5.1	4.1	6.6	5.3
						OK	OK	OK	OK

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Sensitivity: LNT Construction Internal Use

DESIGN CALCULATION OF FOUNDATION FOR ADMIN BUILDING

Load Combination & Calculation of Base pressure

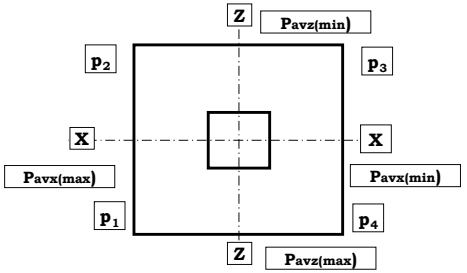
Max. Base pressure = p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)

Min. Base pressure = p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)

Surcharge load = 10 kN/m^2

Net SBC of soil for DL, LL= 350 kN/m^2 For WL/SL case = 437.5 kN/m^2

Gross SBC of soil for DL, LL= 371.6 kN/m^2 For WL/SL case = 459.1 kN/m^2



Allowable Stress Design

Load Combination	Design Load/moments V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)^2.Lz	6.Mx/ Lx.(Lz)^2	Base Pressure (kN/m^2) p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
10 DL	415.02	-10.89	-0.64	184.45	19.36	1.13	204.94	202.68	166.22	163.96	183.34	203.81	165.09	185.58	183.32	O.K
101 DL+T	415.02	-10.89	-0.64	184.45	19.36	1.13	204.94	202.68	166.22	163.96	183.34	203.81	165.09	185.58	183.32	O.K
102 DL+LR	438.53	-11.73	-0.65	194.90	20.85	1.15	216.91	214.60	175.21	172.90	195.31	215.76	174.06	196.06	193.75	O.K
103 D+0.75LR	432.65	-11.52	-0.65	192.29	20.48	1.15	213.92	211.62	172.96	170.66	192.32	212.77	171.81	193.44	191.14	O.K
104 D+WX1	415.02	-10.89	-0.64	184.45	19.36	1.13	204.94	202.68	166.22	163.96	183.34	203.81	165.09	185.58	183.32	O.K
105 D+WX2	415.02	-10.89	-0.64	184.45	19.36	1.13	204.94	202.68	166.22	163.96	183.34	203.81	165.09	185.58	183.32	O.K
106 D+WZ1	415.02	-10.89	-0.64	184.45	19.36	1.13	204.94	202.68	166.22	163.96	183.34	203.81	165.09	185.58	183.32	O.K
107 D+WZ2	415.02	-10.89	-0.64	184.45	19.36	1.13	204.94	202.68	166.22	163.96	183.34	203.81	165.09	185.58	183.32	O.K
108 D+0.7EZ	416.05	-10.87	-51.26	184.91	19.32	91.12	295.36	113.11	256.71	74.46	273.76	204.24	165.59	276.04	93.79	O.K
109 D+0.7EX	417.47	37.18	-0.59	185.54	66.10	1.04	252.68	250.60	120.48	118.40	231.08	251.64	119.44	186.58	184.50	O.K
110 D+0.75WX1+0.75LR	432.65	-11.52	-0.65	192.29	20.48	1.15	213.92	211.62	172.96	170.66	192.32	212.77	171.81	193.44	191.14	O.K
111 D+0.75WX2+0.75LR	432.65	-11.52	-0.65	192.29	20.48	1.15	213.92	211.62	172.96	170.66	192.32	212.77	171.81	193.44	191.14	O.K
112 D+0.75WZ1+0.75LR	432.65	-11.52	-0.65	192.29	20.48	1.15	213.92	211.62	172.96	170.66	192.32	212.77	171.81	193.44	191.14	O.K
113 D+0.75WZ2+0.75LR	432.65	-11.52	-0.65	192.29	20.48	1.15	213.92	211.62	172.96	170.66	192.32	212.77	171.81	193.44	191.14	O.K
114 0.6D+WX1	278.69	-6.53	-0.38	123.86	11.62	0.68	136.16	134.80	112.92	111.57	114.56	135.48	112.25	124.54	123.19	O.K
115 0.6D+WX2	278.69	-6.53	-0.38	123.86	11.62	0.68	136.16	134.80	112.92	111.57	114.56	135.48	112.25	124.54	123.19	O.K
116 0.6D+WZ1	278.69	-6.53	-0.38	123.86	11.62	0.68	136.16	134.80	112.92	111.57	114.56	135.48	112.25	124.54	123.19	O.K
117 0.6D+WZ2	278.69	-6.53	-0.38	123.86	11.62	0.68	136.16	134.80	112.92	111.57	114.56	135.48	112.25	124.54	123.19	O.K
118 0.6D+0.7EX	281.14	41.54	0.33	124.95	73.85	0.59	199.39	198.21	51.70	50.51	177.79	198.80	51.11	125.55	124.36	O.K
119 0.6D+0.7EZ	279.72	-										135.90	112.74	214.99	33.65	O.K

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Strength Design

201 1.4DL	551.35	-15.25	-0.89	245.04	27.10	1.58	273.73	270.57	219.52	216.36	252.13	272.15	217.94	246.63	243.47	
202 1.4D+1.4T+0.5LR	559.01	-16.03	-2.15	248.45	28.50	3.82	280.77	273.14	223.76	216.13	259.17	276.96	219.95	252.27	244.64	
203 1.2D+0.8WX1+1.6LR	520.80	-14.41	-0.78	231.47	25.62	1.39	258.48	255.69	207.24	204.46	236.88	257.09	205.85	232.86	230.08	
204 1.2D+0.8WX2+1.6LR	520.80	-14.41	-0.78	231.47	25.62	1.39	258.48	255.69	207.24	204.46	236.88	257.09	205.85	232.86	230.08	
205 1.2D+0.8WZ1+1.6LR	520.80	-14.41	-0.78	231.47	25.62	1.39	258.48	255.69	207.24	204.46	236.88	257.09	205.85	232.86	230.08	
206 1.2D+0.8WZ2+1.6LR	520.80	-14.41	-0.78	231.47	25.62	1.39	258.48	255.69	207.24	204.46	236.88	257.09	205.85	232.86	230.08	
207 1.2D+1.6WX1+0.5LR	494.94	-13.49	-0.77	219.97	23.98	1.37	245.32	242.58	197.36	194.63	223.72	243.95	196.00	221.34	218.61	
208 1.2D+1.6WZ1+0.5LR	494.94	-13.49	-0.77	219.97	23.98	1.37	245.32	242.58	197.36	194.63	223.72	243.95	196.00	221.34	218.61	
209 1.2D+1.6WX2+0.5LR	494.94	-13.49	-0.77	219.97	23.98	1.37	245.32	242.58	197.36	194.63	223.72	243.95	196.00	221.34	218.61	
210 1.2D+1.6WZ2+0.5LR	494.94	-13.49	-0.77	219.97	23.98	1.37	245.32	242.58	197.36	194.63	223.72	243.95	196.00	221.34	218.61	
211 1.2D+EX	486.68	55.61	-0.69	216.30	98.86	1.23	316.39	313.93	118.68	116.22	294.79	315.16	117.45	217.54	215.08	
212 1.2D+EZ	484.65	-13.04	-73.08	215.40	23.18	129.92	368.50	108.66	322.14	62.30	346.90	238.58	192.22	345.32	85.48	
213 0.9D+1.6WX1	380.94	-9.80	-0.57	169.31	17.42	1.02	187.75	185.71	152.90	150.87	166.15	186.73	151.89	170.33	168.29	
214 0.9D+1.6WZ1	380.94	-9.80	-0.57	169.31	17.42	1.02	187.75	185.71	152.90	150.87	166.15	186.73	151.89	170.33	168.29	
215 0.9D+1.6WX2	380.94	-9.80	-0.57	169.31	17.42	1.02	187.75	185.71	152.90	150.87	166.15	186.73	151.89	170.33	168.29	
216 0.9D+1.6WZ2	380.94	-9.80	-0.57	169.31	17.42	1.02	187.75	185.71	152.90	150.87	166.15	186.73	151.89	170.33	168.29	
217 0.9D+EZ	382.41	-9.77	-72.89	169.96	17.37	129.58	316.91	57.75	282.17	23.01	295.31	187.33	152.59	299.54	40.38	
218 0.9D+EX	384.43	58.87	-0.50	170.86	104.66	0.89	276.41	274.63	67.09	65.31	254.81	275.52	66.20	171.75	169.97	
Max :	559	59	0													
Min :	279	-16	-73													

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 195.31 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 205.31 kN/m^2

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 273.76 kN/m^2

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 283.76 kN/m^2

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Sample calculation of values of base pressure:

Load case : 101 DL+T

Maximum pressure, p_1 (pmax) = 184.45+19.36+1.13= 204.94 kN/m^2

p_2= 184.45+19.36-1.13= 202.68 kN/m^2

p_3= 184.45-(19.36+1.13)= 163.96 kN/m^2

p_4= 184.45-19.36+1.13= 166.22 kN/m^2

Maxm pressure excluding weight of soil (p_{max}-γh)= 204.94-1.2*18= 183.34 kN/m^2

Maximum average Pressure along X, $p_{avx(max)}$ = (204.94+202.68)/2= 203.81 kN/m^2

Minimum average Pressure along X, $p_{avx(minx)}$ = (163.96+166.22)/2= 165.09 kN/m^2

Maximum average Pressure along Y, $p_{avy(max)}$ = (204.94+166.22)/2= 185.58 kN/m^2

Minimum average Pressure along Y, $p_{avy(minx)}$ = (163.96+202.68)/2= 183.32 kN/m^2

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DESIGN CALCULATION OF FOUNDATION FOR ADMIN BUILDING

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.3+2*0.5*0.326)+(0.3+2*0.5*0.326))*2

= 2.504 m

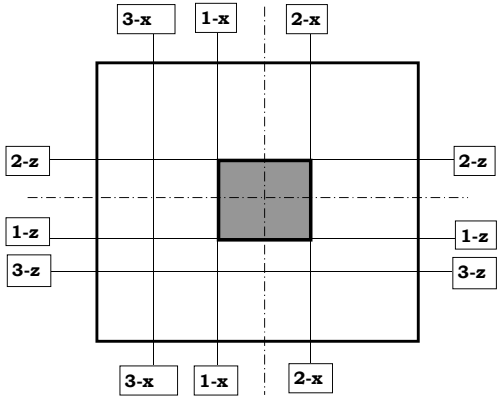
Punching Area= ((0.3+2*0.5*0.326)*(0.3+2*0.5*0.326))

= 0.39 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.4*24+1.3*18)

= 33.00 kN/m^2

Area of footing, A_f= 2.25 m^2



Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	$B.M_{1x}$ kNm	$B.M_{2x}$ kNm	$S.F_{3x}$ kN	$S.F_{(punching)}$ kN
Ultimate Conditions							
201 1.4DL	250.47	239.62	262.83	41.75	34.59	60.50	394.40
202 1.4D+1.4T+0.5LR	254.16	242.75	267.15	42.54	35.02	61.68	400.73
203 1.2D+0.8WX1+1.6LR	236.59	226.35	248.28	39.11	32.34	56.68	369.15
204 1.2D+0.8WX2+1.6LR	236.59	226.35	248.28	39.11	32.34	56.68	369.15
205 1.2D+0.8WZ1+1.6LR	236.59	226.35	248.28	39.11	32.34	56.68	369.15
206 1.2D+0.8WZ2+1.6LR	236.59	226.35	248.28	39.11	32.34	56.68	369.15
207 1.2D+1.6WX1+0.5LR	224.77	215.18	235.70	36.82	30.49	53.36	347.77
208 1.2D+1.6WZ1+0.5LR	224.77	215.18	235.70	36.82	30.49	53.36	347.77
209 1.2D+1.6WX2+0.5LR	224.77	215.18	235.70	36.82	30.49	53.36	347.77
210 1.2D+1.6WZ2+0.5LR	224.77	215.18	235.70	36.82	30.49	53.36	347.77
211 1.2D+EX	236.08	196.53	281.15	46.04	19.95	68.41	340.94
212 1.2D+EZ	220.04	210.76	230.61	35.89	29.77	52.01	339.26
213 0.9D+1.6WX1	172.79	165.83	180.74	26.84	22.24	38.89	253.53
214 0.9D+1.6WZ1	172.79	165.83	180.74	26.84	22.24	38.89	253.53
215 0.9D+1.6WX2	172.79	165.83	180.74	26.84	22.24	38.89	253.53
216 0.9D+1.6WZ2	172.79	165.83	180.74	26.84	22.24	38.89	253.53
217 0.9D+EZ	173.43	166.49	181.35	26.95	22.36	39.05	254.74

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Sample calculation of BM & SF

Load case : 201 1.4DL

p_{1x} = 272.15-(272.15-217.94)*0.6/1.5 = 250.47 kN/m^2 Overhang = 0.6 m

p_{2x} = 217.94+(272.15-217.94)*0.6/1.5 = 239.62 kN/m^2 Effective depth = 0.342 m

p_{3x} = 272.15-(272.15-217.94)*(0.6-0.342)/1.5 = 262.83 kN/m^2

BM_{1x} = (250.47-33)*0.6*0.6/2+0.5*(272.15-250.47)*0.6*0.6*0.67 = 41.75 kN-m/m

BM_{2x} = (239.62-33)*0.6*0.6/2-0.5*(-217.94+239.62)*0.6*0.6*0.67 = 34.59 kN-m/m

SF_{3x} = (262.83-33)*(0.6-0.342)+0.5*(272.15-262.83)*(0.6-0.342) = 60.50 kN/m

$S.F_{(punching)}$ = (2.25-0.39)*(245.04-33) = 394.40 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	46.04	0.39	0.52	0.001	431.24	615.60	16ϕT	326.61	300	200	1005.31
At Top	0.00	0.00	0.00	0.000	0.00	615.60	16ϕT	326.61	300	200	1005.31

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 68.41 kN/m

Shear stress , τ_v= (68.41*1000)/(1000*342)

= 0.20 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 400.73 kN

Strength of Concrete in shear Vc1 = 2533.17 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 1997.67 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 1688.78 kN **Table 22.6.5.2_ (c)**

Design Vc= 1688.78 kN

Allowable shear stress, (ϕ Vc)/bd = 1.48 N/mm^2

Design punching shear stress, τ = (400.73*1000)/(2504*342)

= 0.47 N/mm^2 **No shear reinforcement reqd.**

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Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	$B.M_{1z}$ kNm	$B.M_{2z}$ kNm	$S.F_{3z}$ kN
Ultimate Conditions						
201 1.4DL	245.37	244.73	246.05	38.38	37.96	58.46
202 1.4D+1.4T+0.5LR	249.22	247.69	250.88	39.29	38.28	59.89
203 1.2D+0.8WX1+1.6LR	231.75	231.19	232.35	35.91	35.54	54.69
204 1.2D+0.8WX2+1.6LR	231.75	231.19	232.35	35.91	35.54	54.69
205 1.2D+0.8WZ1+1.6LR	231.75	231.19	232.35	35.91	35.54	54.69
206 1.2D+0.8WZ2+1.6LR	231.75	231.19	232.35	35.91	35.54	54.69
207 1.2D+1.6WX1+0.5LR	220.25	219.70	220.84	33.84	33.47	51.54
208 1.2D+1.6WZ1+0.5LR	220.25	219.70	220.84	33.84	33.47	51.54
209 1.2D+1.6WX2+0.5LR	220.25	219.70	220.84	33.84	33.47	51.54
210 1.2D+1.6WZ2+0.5LR	220.25	219.70	220.84	33.84	33.47	51.54
211 1.2D+EX	216.56	216.06	217.09	33.16	32.83	50.50
212 1.2D+EZ	241.38	189.42	297.86	50.04	15.62	79.07
213 0.9D+1.6WX1	169.51	169.11	169.96	24.67	24.40	37.58
214 0.9D+1.6WZ1	169.51	169.11	169.96	24.67	24.40	37.58
215 0.9D+1.6WX2	169.51	169.11	169.96	24.67	24.40	37.58
216 0.9D+1.6WZ2	169.51	169.11	169.96	24.67	24.40	37.58
217 0.9D+EZ	195.88	144.04	252.20	41.82	7.49	66.55
218 0.9D+EX	171.04	170.68	171.42	24.93	24.70	37.97

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Sample calculation of BM & SF

Load case : 201 1.4DL

p_{1z} = 246.63-(246.63-243.47)*0.6/1.5= 245.37 kN/m^2

p_{2z} = 243.47+(246.63-243.47)*0.6/1.5= 244.73 kN/m^2

p_{3z} = 246.63-(246.63-243.47)*(0.6-0.326)/1.5= 246.05 kN/m^2

Overhang = 0.6 m

Effective depth = 0.326 m

BM_{1z} = (245.37-33)*0.6*0.6/2+0.5*(246.63-245.37)*0.6*0.6*0.67 = 38.38 kN-m/m

BM_{2z} = (244.73-33)*0.6*0.6/2-0.5*(-243.47+244.73)*0.6*0.6*0.67 = 37.96 kN-m/m

SF_{3z} = (246.05-33)*(0.6-0.326)+0.5*(246.63-246.05)*(0.6-0.326) = 58.46 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	50.04	0.471	0.628	0.0015	516.76	586.80	16ϕT	342.64	300	200	1005.3
Top	0.00	0.000	0.000	0.0000	0.00	586.80	16ϕT	342.64	300	200	1005.3

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 79.07 kN/m

Shear stress , τ_v= (79.07*1000)/(1000*326)

= 0.24 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :			
At Bottom	**16ϕT**	**200**	c/c
At Top	**16ϕT**	**200**	c/c
Along Z Direction :			
At Bottom	**16ϕT**	**200**	c/c
At Top	**16ϕT**	**200**	c/c

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REFERENCES

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) clause 4.1.

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ATTACHMENT 1 STAAD INPUT DATA

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STAAD INPUT:

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 12-Oct-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 0 0; 2 13.35 0 0; 3 0 0 19.15; 4 13.35 0 19.15; 5 0 3.05 0; 6 13.35 3.05 0;
7 0 3.05 19.15; 8 13.35 3.05 19.15; 9 0 -1.5 0; 10 13.35 -1.5 0;
11 0 -1.5 19.15; 12 13.35 -1.5 19.15; 13 6 -1.5 0; 14 6 0 0; 15 6 3.05 0;
16 6 -1.5 19.15; 17 6 0 19.15; 18 6 3.05 19.15; 19 0 -1.5 3.55; 20 0 0 3.55;
21 0 3.05 3.55; 22 13.35 -1.5 3.55; 23 13.35 0 3.55; 24 13.35 3.05 3.55;
25 6 -1.5 3.55; 26 6 0 3.55; 27 6 3.05 3.55; 28 0 -1.5 9.7; 29 0 0 9.7;
30 0 3.05 9.7; 31 13.35 -1.5 9.7; 32 13.35 0 9.7; 33 13.35 3.05 9.7;
34 6 -1.5 9.7; 35 6 0 9.7; 36 6 3.05 9.7; 37 0 -1.5 15.4; 38 0 0 15.4;
39 0 3.05 15.4; 40 13.35 -1.5 15.4; 41 13.35 0 15.4; 42 13.35 3.05 15.4;
43 6 -1.5 15.4; 44 6 0 15.4; 45 6 3.05 15.4; 46 2.35 0 0; 50 4.325 0 19.15;
52 2.35 3.05 0; 53 2.35 3.05 3.55; 54 2.35 3.05 9.7; 55 2.35 3.05 15.4;
56 4.325 3.05 15.4; 57 4.325 3.05 19.15; 58 9.675 3.05 0; 59 9.675 3.05 3.55;
60 9.675 3.05 9.7; 61 13.35 3.05 6.05; 62 6 3.05 6.05; 65 9.675 0 0;
71 13.35 0 6.05; 72 6 3.05 13.45;
MEMBER INCIDENCES
1 9 1; 2 1 5; 3 11 3; 4 3 7; 5 12 4; 6 4 8; 7 10 2; 8 2 6; 9 5 52; 10 7 57;
11 5 21; 12 6 24; 13 1 46; 14 3 50; 15 2 23; 16 1 20; 17 14 65; 18 13 14;
19 15 58; 20 14 15; 21 17 4; 22 16 17; 23 18 8; 24 17 18; 25 20 29; 26 19 20;
27 21 30; 28 20 21; 29 23 71; 30 22 23; 31 24 61; 32 23 24; 33 25 26; 34 26 27;
35 29 38; 36 28 29; 37 30 39; 38 29 30; 39 32 41; 40 31 32; 41 33 42; 42 32 33;
43 34 35; 44 35 36; 45 38 3; 46 37 38; 47 39 7; 48 38 39; 49 41 4; 50 40 41;
51 42 8; 52 41 42; 53 43 44; 54 44 45; 55 21 53; 56 27 59; 57 30 54; 58 36 60;
59 39 55; 60 45 42; 61 15 27; 62 27 62; 63 36 72; 64 45 18; 75 46 14; 79 50 17;
85 52 15; 86 53 27; 88 54 36; 90 55 56; 92 56 45; 93 57 18; 95 58 6; 96 59 24;
97 60 33; 100 61 33; 101 62 36; 107 65 2; 117 71 32; 119 72 45;
DEFINE MATERIAL START
ISOTROPIC CONCRETE_M35
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
G 9.28139e+006
TYPE CONCRETE
STRENGTH FCU 28000
END DEFINE MATERIAL
CONSTANTS
MATERIAL CONCRETE_M35 ALL
MEMBER PROPERTY AMERICAN
1 TO 8 18 20 22 24 53 54 PRIS YD 0.4 ZD 0.3
26 28 30 32 TO 34 36 38 40 42 TO 44 46 48 50 52 PRIS YD 0.3 ZD 0.4
9 TO 12 19 23 27 31 37 41 47 51 55 TO 64 85 86 88 90 92 93 95 TO 97 100 101 -
119 PRIS YD 0.65 ZD 0.3
13 TO 17 21 25 29 35 39 45 49 75 79 107 117 PRIS YD 0.7 ZD 0.3
SUPPORTS
9 TO 13 16 19 22 25 28 31 34 37 40 43 FIXED
```

```
MEMBER CRACKED
1 TO 8 18 20 22 24 26 28 30 32 TO 34 36 38 40 42 TO 44 46 48 50 52 TO 53 -
54 REDUCTION RIX 0.700000 RIY 0.700000 RIZ 0.700000
9 TO 17 19 21 23 25 27 29 31 35 37 39 41 45 47 49 51 55 TO 64 75 79 85 86 -
88 90 92 93 95 TO 97 100 101 107 117 -
119 REDUCTION RIX 0.350000 RIY 0.350000 RIZ 0.350000
DEFINE IBC 2003
SDS 0.106 SD1 0.032 S1 0.02 IE 1 RX 2.5 RZ 2.5 SCLASS 4
SELFWEIGHT 1
MEMBER WEIGHT
*Vertical
MEMBER WEIGHT
9 TO 12 19 23 27 31 37 41 47 51 85 93 95 100 UNI GY -7.8
9 19 85 95 UNI GY -10.4
55 56 86 96 UNI GY -28.3
57 58 88 97 UNI GY -34.5
59 60 90 92 UNI GY -27.5
10 23 93 UNI GY -11
13 TO 17 21 25 29 35 39 45 49 75 79 107 117 UNI GY -12.2
*X-direction
MEMBER WEIGHT
9 TO 12 19 23 27 31 37 41 47 51 85 93 95 100 UNI GX 7.8
9 19 85 95 UNI GX 10.4
55 56 86 96 UNI GX 28.3
57 58 88 97 UNI GX 34.5
59 60 90 92 UNI GX 27.5
10 23 93 UNI GX 11
13 TO 17 21 25 29 35 39 45 49 75 79 107 117 UNI GX 12.2
*Z-direction
MEMBER WEIGHT
9 TO 12 19 23 27 31 37 41 47 51 85 93 95 100 UNI GZ 7.8
9 19 85 95 UNI GZ 10.4
55 56 86 96 UNI GZ 28.3
57 58 88 97 UNI GZ 34.5
59 60 90 92 UNI GZ 27.5
10 23 93 UNI GZ 11
13 TO 17 21 25 29 35 39 45 49 75 79 107 117 UNI GZ 12.2
DEFINE WIND LOAD
TYPE 1 WIND X WINDWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 1 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
2 5.000 M 13.500 M 20.000 M 0.800 0.010 0 -
0 0 0 0 0.590 1.000 1.000 0.850 0 -
0 0 0 0.858 0.800 -0.180
!> END GENERATED DATA BLOCK
INT 0.630908 0.630908 0.631928 0.632943 0.633953 0.634958 0.635958 0.636953 -
 0.637943 0.638928 0.639908 0.640884 0.641855 0.642822 0.643783 HEIG  0  4.572 -
 4.60492  4.63785  4.67077  4.70369  4.73662  4.76954  4.80246  4.83538 -
 4.86831  4.90123  4.93415  4.96708  5
TYPE 2 WIND X LEEWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 1 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
2 5.000 M 13.000 M 20.000 M 0.800 0.010 1 -
0 0 0 0 0.590 1.000 1.000 0.850 0 -
```

0 0 0 0.858 -0.500 0.180
!> END GENERATED DATA BLOCK
INT -0.452521 -0.452521 0.630908 HEIG 0 5 4.572
TYPE 3 SIDEWARD X
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 1 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
2 5.000 M 13.000 M 20.000 M 0.800 0.010 2 -
0 0 0 0 0.590 1.000 1.000 0.850 0 -
0 0 0 0.858 -0.700 0.180
!> END GENERATED DATA BLOCK
INT -0.580029 -0.580029 HEIG 0 5
TYPE 4 WIND Z WINDWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 1 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
2 5.000 M 20.000 M 13.000 M 0.800 0.010 0 -
0 0 0 0 0.590 1.000 1.000 0.850 0 -
0 0 0 0.869 0.800 -0.180
!> END GENERATED DATA BLOCK
INT 0.637166 0.637166 0.638199 0.639227 0.640249 0.641267 0.642279 0.643287 -
0.644289 0.645287 0.646279 0.647267 0.64825 0.649229 0.650203 HEIG 0 4.572 -
4.60492 4.63785 4.67077 4.70369 4.73662 4.76954 4.80246 4.83538 -
4.86831 4.90123 4.93415 4.96708 5
TYPE 5 WIND Z LEEWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 1 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
2 5.000 M 20.000 M 13.000 M 0.800 0.010 1 -
0 0 0 0 0.590 1.000 1.000 0.850 0 -
0 0 0 0.869 -0.392 0.180
!> END GENERATED DATA BLOCK
INT -0.387011 -0.387011 HEIG 0 5
TYPE 6 SIDEWARD Z
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 1 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
2 5.000 M 20.000 M 13.000 M 0.800 0.010 2 -
0 0 0 0 0.590 1.000 1.000 0.850 0 -
0 0 0 0.869 -0.700 0.180
!> END GENERATED DATA BLOCK
INT -0.585646 -0.585646 HEIG 0 5
LOAD 1 LOADTYPE Seismic TITLE EQK LOAD Z
IBC LOAD Z 1
PERFORM ANALYSIS
CHANGE
LOAD 2 LOADTYPE Seismic TITLE EQK LOAD X
IBC LOAD X 1
PERFORM ANALYSIS
CHANGE
LOAD 3 LOADTYPE Dead TITLE DL
SELFWEIGHT Y -1
MEMBER LOAD
9 TO 12 19 23 27 31 37 41 47 51 85 93 95 100 UNI GY -7.8
9 19 85 95 UNI GY -10.4
55 56 86 96 UNI GY -28.3
57 58 88 97 UNI GY -34.5
59 60 90 92 UNI GY -27.5

10 23 93 UNI GY -11
10 11 23 27 37 47 93 UMOM X 5.8
12 19 31 41 51 85 95 100 UMOM X -5.8
13 TO 17 21 25 29 35 39 45 49 75 79 107 117 UNI GY -12.2
LOAD 4 LOADTYPE Live REDUCIBLE TITLE LR
MEMBER LOAD
9 19 85 95 UNI GY -0.9
55 56 86 96 UNI GY -2.5
57 58 88 97 UNI GY -3
59 60 90 92 UNI GY -2.4
10 23 93 UNI GY -1
9 TO 12 19 23 27 31 37 41 47 51 85 93 95 100 UNI GY -0.3
10 11 23 27 37 47 93 UMOM X 0.09
9 12 19 31 41 51 85 95 100 UMOM X -0.09
LOAD 5 LOADTYPE Wind TITLE WL+X
WIND LOAD X 1 TYPE 1 YR 0 6
WIND LOAD -X 1 TYPE 2 YR 0 6
WIND LOAD -Z 1 TYPE 3 YR 0 6
WIND LOAD -Z -1 TYPE 3 YR 0 6
LOAD 6 LOADTYPE Wind TITLE WL+Z
WIND LOAD Z 1 TYPE 4 YR 0 6
WIND LOAD -Z 1 TYPE 5 YR 0 6
WIND LOAD -X 1 TYPE 6 YR 0 6
WIND LOAD -X -1 TYPE 6 YR 0 6
LOAD 7 LOADTYPE Wind TITLE WL-X
WIND LOAD X -1 TYPE 1 YR 0 6
WIND LOAD -X -1 TYPE 2 YR 0 6
WIND LOAD -Z 1 TYPE 3 YR 0 6
WIND LOAD -Z -1 TYPE 3 YR 0 6
LOAD 8 LOADTYPE Wind TITLE WL-Z
WIND LOAD Z -1 TYPE 4 YR 0 6
WIND LOAD -Z -1 TYPE 5 YR 0 6
WIND LOAD -X 1 TYPE 6 YR 0 6
WIND LOAD -X -1 TYPE 6 YR 0 6
LOAD 9 LOADTYPE Temperature TITLE TEMP
TEMPERATURE LOAD
2 4 6 8 TO 12 19 20 23 24 27 28 31 32 34 37 38 41 42 44 47 48 51 52 -
54 TO 64 85 86 88 90 92 93 95 TO 97 100 101 119 TEMP 25
LOAD COMB 10 DL
3 1.0
LOAD COMB 101 DL+T
3 1.0
LOAD COMB 102 DL+LR
3 1.0 4 1.0
LOAD COMB 103 D+0.75LR
3 1.0 4 0.75
LOAD COMB 104 D+WX1
3 1.0
LOAD COMB 105 D+WX2
3 1.0
LOAD COMB 106 D+WZ1
3 1.0
LOAD COMB 107 D+WZ2
3 1.0

LOAD COMB 108 D+0.7EZ
3 1.0 1 0.7
LOAD COMB 109 D+0.7EX
3 1.0 2 0.7
LOAD COMB 110 D+0.75WX1+0.75LR
3 1.0 4 0.75
LOAD COMB 111 D+0.75WX2+0.75LR
3 1.0 4 0.75
LOAD COMB 112 D+0.75WZ1+0.75LR
3 1.0 4 0.75
LOAD COMB 113 D+0.75WZ2+0.75LR
3 1.0 4 0.75
LOAD COMB 114 0.6D+WX1
3 0.6
LOAD COMB 115 0.6D+WX2
3 0.6
LOAD COMB 116 0.6D+WZ1
3 0.6
LOAD COMB 117 0.6D+WZ2
3 0.6
LOAD COMB 118 0.6D+0.7EX
3 0.6 2 0.7
LOAD COMB 119 0.6D+0.7EZ
3 0.6 1 0.7
LOAD COMB 201 1.4DL
3 1.4
LOAD COMB 202 1.4D+1.4T+0.5LR
3 1.4 9 1.4 4 0.5
LOAD COMB 203 1.2D+0.8WX1+1.6LR
3 1.2 4 1.6
LOAD COMB 204 1.2D+0.8WX2+1.6LR
3 1.2 4 1.6
LOAD COMB 205 1.2D+0.8WZ1+1.6LR
3 1.2 4 1.6
LOAD COMB 206 1.2D+0.8WZ2+1.6LR
3 1.2 4 1.6
LOAD COMB 207 1.2D+1.6WX1+0.5LR
3 1.2 4 0.5
LOAD COMB 208 1.2D+1.6WZ1+0.5LR
3 1.2 4 0.5
LOAD COMB 209 1.2D+1.6WX2+0.5LR
3 1.2 4 0.5
LOAD COMB 210 1.2D+1.6WZ2+0.5LR
3 1.2 4 0.5
LOAD COMB 211 1.2D+EX
3 1.2 2 1.0
LOAD COMB 212 1.2D+EZ
3 1.2 1 1.0
LOAD COMB 213 0.9D+1.6WX1
3 0.9
LOAD COMB 214 0.9D+1.6WZ1
3 0.9
LOAD COMB 215 0.9D+1.6WX2
3 0.9

```
LOAD COMB 216 0.9D+1.6WZ2
3 0.9
LOAD COMB 217 0.9D+EZ
3 0.9 1 1.0
LOAD COMB 218 0.9D+EX
3 0.9 2 1.0
PERFORM ANALYSIS
START CONCRETE DESIGN
CODE ACI
CLB 0.04 ALL
CLS 0.04 ALL
CLT 0.04 ALL
FC 28000 ALL
FYMAIN 414000 ALL
MINSEC 10 ALL
MINMAIN 16 MEMB 9 TO 17 19 21 23 25 27 29 31 35 37 39 41 45 47 49 51 -
55 TO 64 75 79 85 86 88 90 92 93 95 TO 97 100 101 107 117 119
MINMAIN 25 MEMB 1 TO 8 11 12 15 16 18 20 22 24 TO 54 61 TO 64 100 101 117 119
DESIGN BEAM 9 TO 17 19 21 23 25 27 29 31 35 37 39 41 45 47 49 51 55 TO 64 -
75 79 85 86 88 90 92 93 95 TO 97 100 101 107 117 119
DESIGN COLUMN 1 TO 8 18 20 22 24 26 28 30 32 TO 34 36 38 40 42 TO 44 46 48 -
50 52 TO 54
END CONCRETE DESIGN
FINISH
```